

TCRR Phase 1 — Benchmarking Report

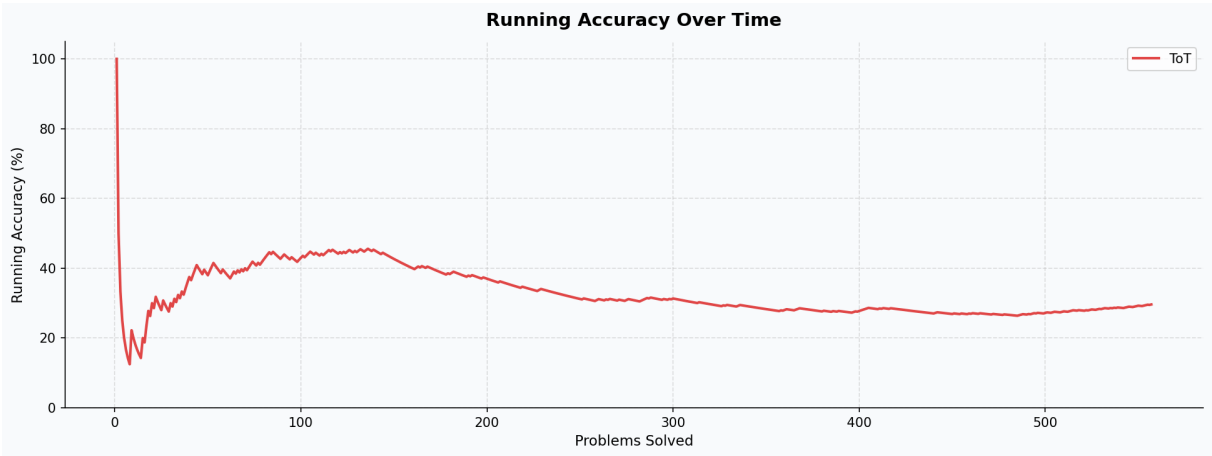
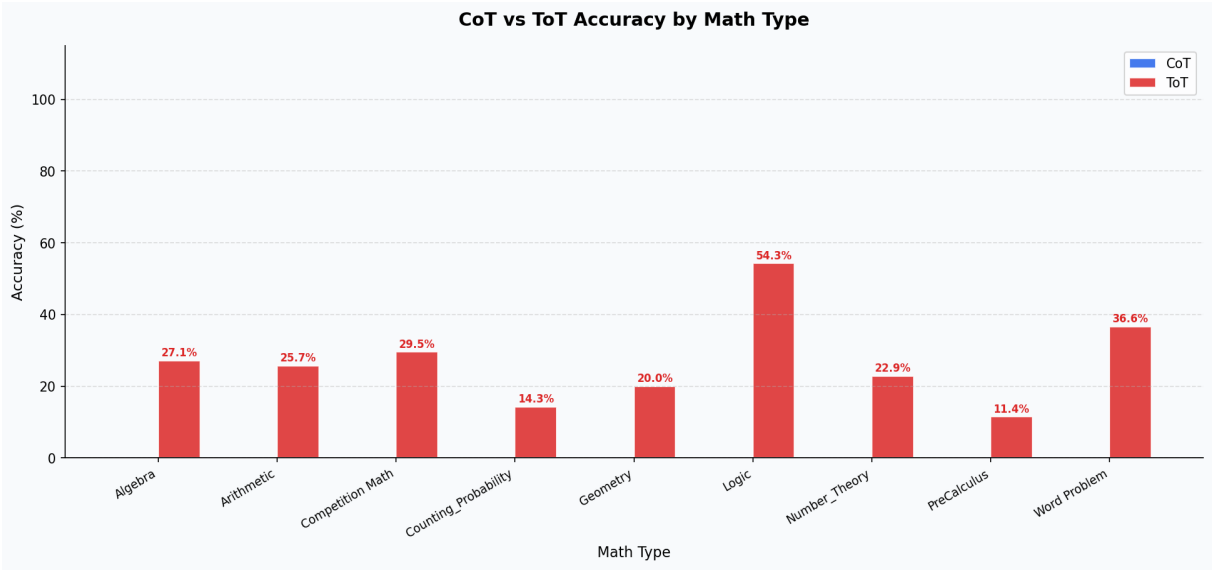
Type-Aware Cost-Efficient Reasoning Router

Questions: 557 | Generated: 2026-03-12 15:45:07 | Model: llama-3.3-70b-versatile (Groq)

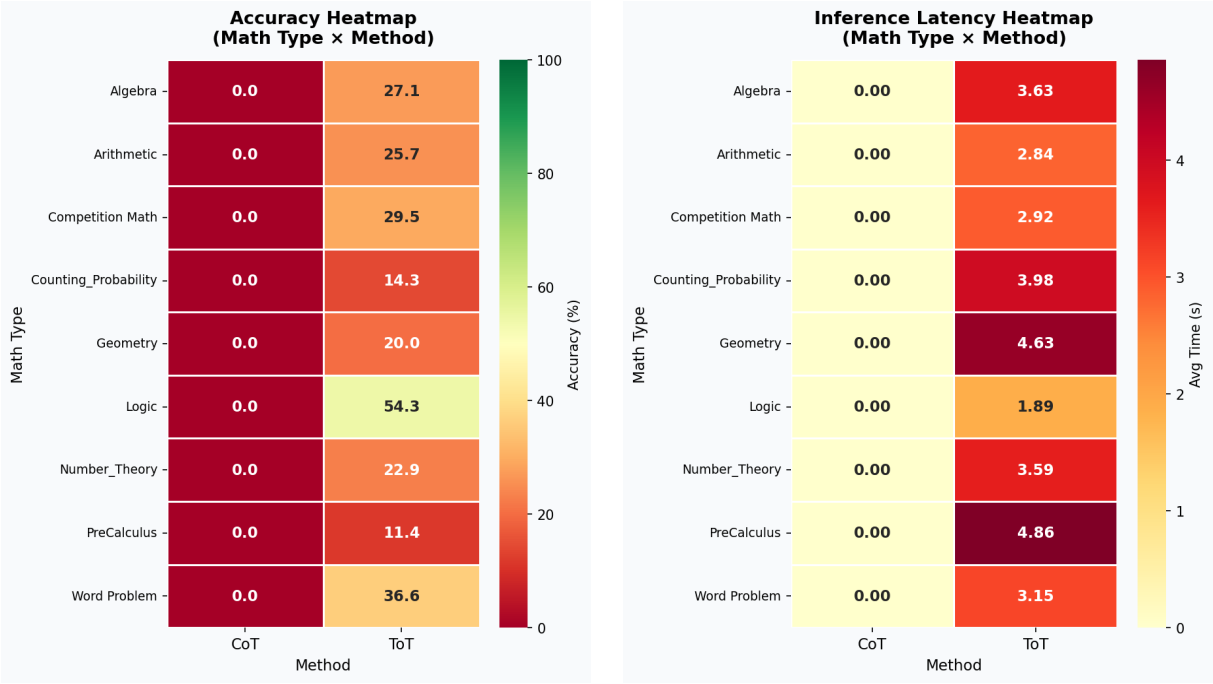
1. Per-Category Summary

Category	n	CoT Acc%	ToT Acc%	CoT Lat(s)	ToT Lat(s)	CoT Tokens	ToT Tokens	CoT Steps	ToT Steps	Winner
Algebra	70	—	27.1	—	3.63	—	2543.0	—	7.0	—
Arithmetic	35	—	25.7	—	2.84	—	2048.0	—	7.9	—
Competition Math	105	—	29.5	—	2.92	—	2105.0	—	7.6	—
Counting_Probability	35	—	14.3	—	3.98	—	2426.0	—	7.7	—
Geometry	35	—	20.0	—	4.63	—	2747.0	—	7.0	—
Logic	35	—	54.3	—	1.89	—	1406.0	—	8.0	—
Number_Theory	35	—	22.9	—	3.59	—	2364.0	—	7.5	—
PreCalculus	35	—	11.4	—	4.86	—	2794.0	—	6.2	—
Word Problem	172	—	36.6	—	3.15	—	1765.0	—	7.7	—
Overall	0/557	—	29.6	—	3.35	—	2128.0	—	—	—

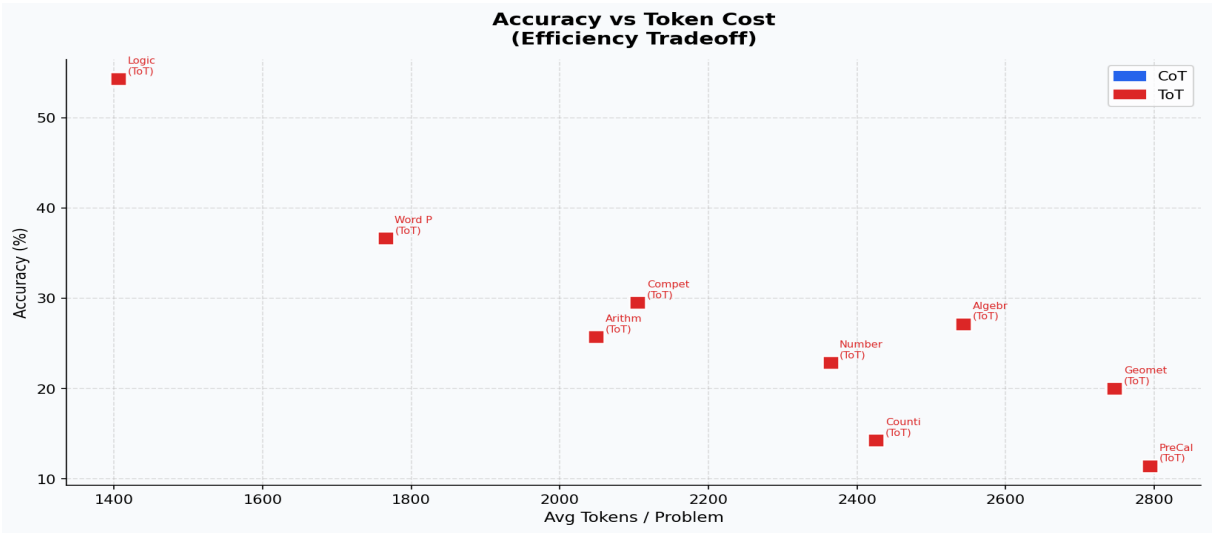
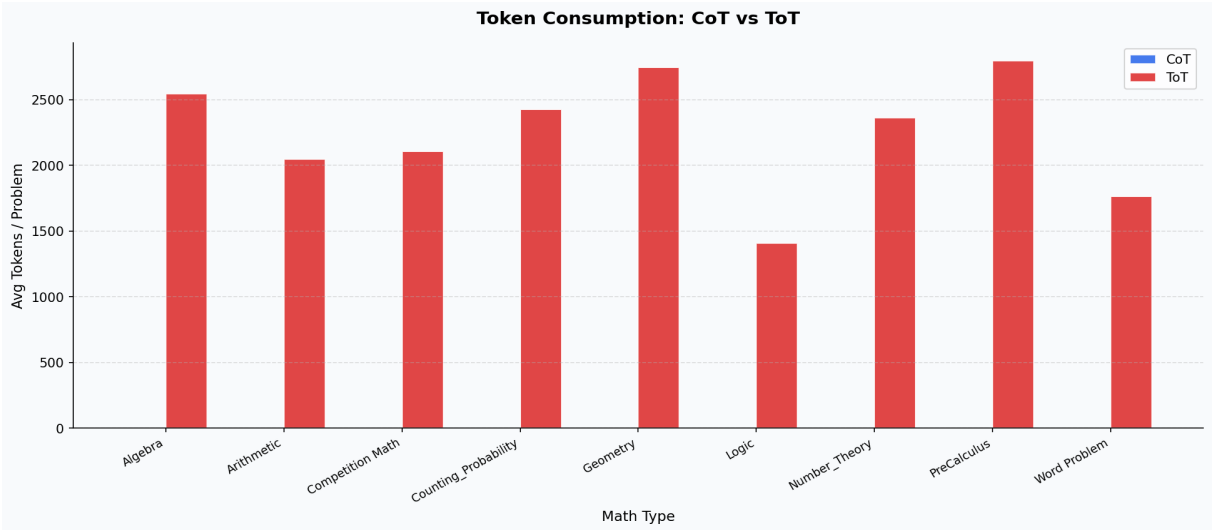
2. Accuracy Analysis



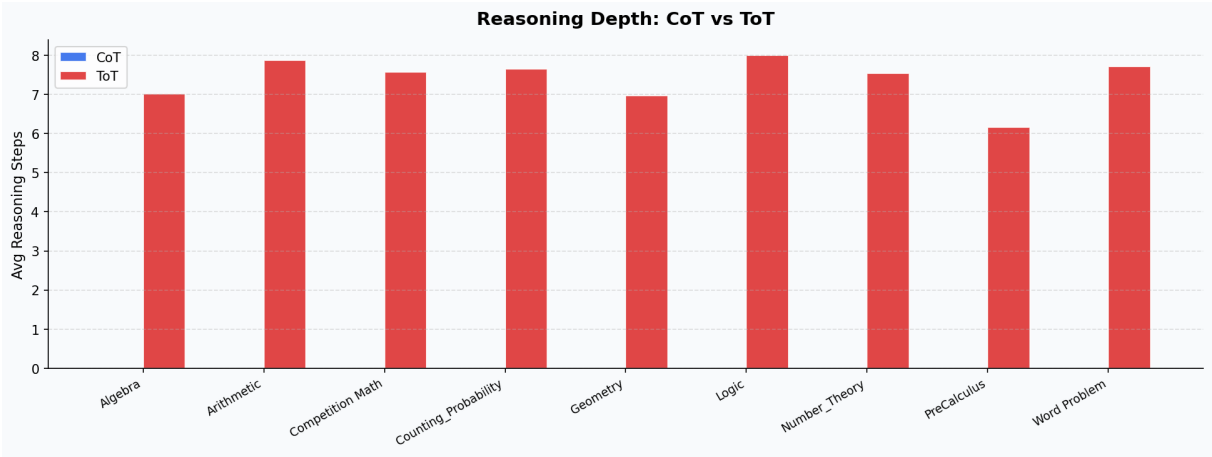
3. Heatmaps



4. Token Cost & Efficiency



5. Reasoning Depth & Strategy



6. Router Recommendation (Phase 2)

Math Type	CoT Acc%	ToT Acc%	Recommended	Token Δ	Reason
Algebra	—	—	Need data	—	—
Arithmetic	—	—	Need data	—	—
Competition Math	—	—	Need data	—	—
Counting_Probability	—	—	Need data	—	—
Geometry	—	—	Need data	—	—
Logic	—	—	Need data	—	—
Number_Theory	—	—	Need data	—	—
PreCalculus	—	—	Need data	—	—
Word Problem	—	—	Need data	—	—